



Digital Currencies – Types & Future

Learning Objectives

- Define digital currency
- Identify types of digital currencies
- Compare advantages and risks
- Explain regulatory challenges
- Discuss future trends

What is Digital Currency?

- Currency that exists only in **digital form**
- No physical notes or coins
- Used through computers, mobile devices, and networks
- Depends on **internet and digital infrastructure**
- **Examples:**
Digital wallets, online payments, cryptocurrencies
- Money in **eSewa wallet** is digital money
But it is **not a digital currency**

Why eSewa Is Digital Money but Not a Digital Currency?

- Just because something is digital does not mean it becomes a new currency.
- Online PDF of a book ≠ new book
- Digital photo of citizenship ≠ new citizenship
-  **Same thing, different form**

Why eSewa Is Digital Money but Not a Digital Currency?

- “Money inside eSewa is not new money. It is still Nepali Rupee.”
- NPR is issued by **Nepal Rastra Bank**
- You deposit NPR into bank
- eSewa only **stores and transfers** that NPR digitally
-  eSewa = **digital storage & transfer system**, not a currency creator

Why eSewa Is Digital Money but Not a Digital Currency?

- “Can eSewa print money?”

“If it cannot create money, it is not a digital currency.”

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Why eSewa Is Digital Money but Not a Digital Currency?

Digital Money

- Existing national currency
- Just in electronic form
- Controlled by central bank
- Example: NPR in eSewa, bank app

Digital Currency

- New form of money
- Has its own system
- May or may not be issued by government
- Example: Bitcoin, CBDC

Why eSewa Is Digital Money but Not a Digital Currency?

Question	eSewa Money	Cryptocurrency
Issued by	Nepal Rastra Bank	No one
Controlled by	Government	Network
Blockchain	No	Yes
Legal tender	Yes	No (Nepal)
Value	Fixed (NPR)	Changes daily

Why eSewa Is NOT a Digital Currency

- “Digital currency means the currency itself is digital by design.”
- But:
- NPR was **designed as physical currency**
- eSewa just digitizes it
- Analogy:
- Scanning a paper document ≠ digital-born document
-  So:
- “Digital money ≠ Digital currency”

Why eSewa Is NOT a Digital Currency

- “eSewa does not change the **nature** of money, only the **form**.”
- All cryptocurrency is digital
- All digital currency is digital
- But **not all digital money is digital currency**
- eSewa money is digital **representation** of NPR, not a new currency

Types of Digital Currencies

- Digital currencies are mainly of **three types**:
- Cryptocurrencies
- Stablecoins
- Central Bank Digital Currency (CBDC)

Cryptocurrency – Definition

- Internet money
- No government control
- Runs on blockchain
- Anyone can send money directly to anyone
- Secured using cryptography (secret writing)
- **Peer-to-peer system**

“Digital cash without Nepal Rastra Bank”

Types of Cryptocurrency

Payment Cryptocurrencies (Used as Digital Money)

Aspect	Explanation
Purpose	Digital money for sending and receiving payments
How it works	Peer-to-peer, no bank in between
Examples	Bitcoin (BTC), Litecoin (LTC)
If Allowed in Nepal	Cheaper remittance, direct international transfer
Major Risks	High price fluctuation, weak user protection
Main Goal	Replace traditional money

Platform / Smart Contract Cryptocurrencies

Aspect

Explanation

Purpose

Run applications and smart contracts

How it works

Blockchain acts as a programmable platform

Examples

Ethereum (ETH), Solana (SOL)

If Allowed in Nepal

Nepali developers could build blockchain apps

Main Use

Technology, not daily payments

Key Risk

Complex system, high technical barrier

Example: Bitcoin

- Bitcoin is:
- Like **gold**, but digital
- Limited supply
- Price changes a lot
- **If Bitcoin were allowed in Nepal:**
- People could send money abroad easily
- Traders could accept global payments
- But prices could crash suddenly

What Cryptocurrency Could Do in Nepal

If allowed:

- Faster foreign payments
- Freelancers paid directly from abroad
- Less dependence on banks

But:

- NRB cannot control it
- High risk for common people

Why Cryptocurrency Is Banned in Nepal

- Difficult to track transactions
- Risk of money laundering
- High fraud potential
- No consumer protection
- **Key Line for Exam:**
“Nepal bans cryptocurrency due to financial and regulatory risks.”

Cryptocurrency – Key Features

- Decentralized
- Transparent public ledger
- Limited supply
- Global accessibility
- Highly volatile

Examples of Cryptocurrencies

- Bitcoin (BTC)
- Ethereum (ETH)
- Binance Coin (BNB)
- Ripple (XRP)

Advantages of Cryptocurrencies

- No central authority
- Low transaction cost
- Fast cross-border payments
- High transparency
- Resistant to censorship

Risks of Cryptocurrencies

- Extreme price volatility
- Used in illegal activities
- Cybersecurity risks
- No legal protection in many countries
- Loss of private keys = loss of money

Stablecoin – Definition

- Digital currency designed to reduce volatility
- Value does NOT change much
- Linked to US Dollar
- Bridge between crypto and traditional money

Types of Stablecoins

- Fiat-backed (USD, EUR)
- Crypto-backed
- Algorithmic stablecoins
- **Most common:** Fiat-backed

Examples of Stablecoins

- Tether (USDT)
- USD Coin (USDC)
- Binance USD (BUSD)

Advantages of Stablecoins

- Price stability
- Faster digital payments
- Useful for remittance
- Reduces crypto market risk

Risks of Stablecoins

- Dependence on reserve trust
- Centralized control
- Regulatory uncertainty
- Risk of reserve mismanagement

CBDC – Definition

- **Central Bank Digital Currency (CBDC):**
- **CBDC = Digital Nepali Rupee**
- Issued by Nepal Rastra Bank
- Same value as cash
- Legal money

“eSewa-like app, but owned by NRB”

Example: Digital Rupee (Nepal Scenario)

- If Nepal launches Digital Rupee:
- People pay directly using NRB wallet
- No need for middle companies
- Government controls security

What CBDC Could Do in Nepal

- Reduce fake notes
- Improve tax collection
- Help rural people access money
- Reduce cash printing cost

Possible Problems of CBDC in Nepal

- Privacy concerns
- Cybersecurity risk
- Internet dependency
- Fear of too much government control

Types of CBDC

- Retail CBDC – for public use
- Wholesale CBDC – for banks and institutions

Examples of CBDC

- Digital Yuan – China
- eNaira – Nigeria
- Digital Rupee – India (pilot)

Advantages of CBDC

- Government-backed and stable
- Reduces cash handling cost
- Improves monetary policy
- Promotes financial inclusion
- Increases transaction transparency

Risks of CBDC

- Privacy concerns
- Cybersecurity threats
- High infrastructure cost
- Excessive government control

Comparison Overview

Feature	Crypto	Stablecoin	CBDC
Control	Decentralized	Semi-centralized	Centralized
Stability	Low	High	Very High
Legal Status	Uncertain	Limited	Legal Tender

General Advantages of Digital Currencies

- Faster transactions
- Lower operational costs
- Financial inclusion
- Transparency
- 24/7 availability

General Risks of Digital Currencies

- Cybercrime
- Data privacy issues
- Technical complexity
- Lack of awareness
- Regulatory uncertainty

Regulatory Challenges

- No global regulatory framework
- Money laundering risks
- Terrorist financing
- Taxation difficulties
- Consumer protection issues

Government Approaches

- Complete ban
- Partial regulation
- Regulatory sandbox
- Development of CBDCs

Future of Digital Currencies

- Decline of physical cash
- Growth of CBDCs
- Integration with AI & IoT
- Smart contracts
- Asset tokenization

Digital Currencies & Future Economy

- Supports digital trade
- Enables global transactions
- Encourages fintech innovation
- Strengthens digital financial systems